Python Fundamentals

A concise reference to Python syntax and programming patterns used in data and AI applications.

What Python is

Python is a general-purpose, interpreted programming language. It is popular in automation, data analysis, machine learning, and back-end development because its syntax is readable and its ecosystem includes many libraries. Python source files typically use the .py extension.

Variables and core data types

A variable is a name that refers to a value. Python does not require a type declaration when assigning a value. Common built-in types include int (whole number), float (decimal number), str (text), bool (True or False), and NoneType (the value None).

```
count = 3
price = 9.99
project = "RAG demo"
is_ready = True
```

Use type(value) to inspect a type. Strings are immutable; methods such as .lower() return a new string rather than modifying the original.

Lists, tuples, dictionaries, and sets

- List: ordered, mutable sequence. Example: `topics = ["python", "rag"]`. Access the first item with `topics[0]` and add an item with `topics.append("ml")`.
- Tuple: ordered, immutable sequence. Example: `point = (10, 20)`. Choose a tuple when values should not be reassigned by position.
- Dictionary: mapping from keys to values. Example: `person = {"name": "Maya", "role": "analyst"}`. Read the role with `person["role"]`.
- Set: mutable collection of unique hashable elements. Example: `tags = {"ai", "python"}`. Useful for removing duplicates and membership checks.

Control flow and loops

Use if, elif, and else for conditional branches. Use for to iterate over a sequence, and while to repeat while a condition remains true. Python uses indentation to define code blocks.

```
scores = [0.91, 0.52, 0.83]
for index, score in enumerate(scores):
    if score >= 0.80:
        print(index, score)
```

enumerate(items) yields pairs of (index, item). zip(a, b) pairs elements from two iterables and stops at the shortest by default. range(len(items)) generates integer indexes from 0 through len(items) - 1.

Functions and return values

Define a function with def. Arguments are inputs; return supplies the output. A function that reaches its end without a return statement returns None.

```
def convert_distance(distance):
    return 1 - distance

result = convert_distance(0.25)  # 0.75
```

Errors and working with files

Catch expected exceptions with try/except, ideally naming the specific error class. A context manager such as with open(path, "r", encoding="utf-8") as file closes a file automatically. Do not hide errors with a broad except: unless there is a clear reason.

For project dependencies, create a virtual environment with python -m venv .venv and install packages using python -m pip install package_name. A requirements file records packages needed to reproduce an environment.

Practical connection to RAG

Python applications often use loops to process search results and dictionaries to store retrieved document fields. A typical result dictionary contains an ID, text content, source metadata, a distance, and a rank. Python itself does not measure semantic similarity; a vector search library returns distances or scores.

Questions this document can answer

Q: How is a list different from a tuple?
A: A list is mutable, while a tuple is immutable; both preserve order.

Q: What does enumerate() do?
A: It yields the index and value of each element during iteration.

Q: Why use a dictionary in a retrieval pipeline?
A: A dictionary can attach named fields such as document ID, content, metadata, and score to a single retrieved result.